

HAYS, KS, USA

WMO: 724518

Lat: 38.850N

Lon: 99.267W

Elev: 1998

StdP: 13.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	7.9	-6.5	4.2	7.6	-1.7	5.4	11.3	32.4	41.7	28.4	36.0	8.9	340	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	100.9	71.4	98.8	71.4	94.9	71.0	75.5	90.8	74.3	90.2	73.0	89.3	16.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	126.7	82.6	70.1	119.4	81.0	68.3	112.1	79.7	40.5	91.3	39.1	90.0	37.9	89.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.7	26.5	23.9	DB	-4.0	106.4	4.1	2.9	-7.0	108.4	-9.4	110.1	-11.7	111.7	-14.7	113.8	
			WB	-4.8	77.9	3.8	1.5	-7.5	79.0	-9.8	79.9	-11.9	80.8	-14.7	81.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	30.4	33.4	43.4	52.6	63.2	74.5	79.5	77.3	69.2	55.4	42.5	31.5
	DBStd	19.79	10.75	11.83	11.11	9.69	8.71	7.38	6.29	6.06	8.78	9.57	9.46	10.04
	HDD50	2315	608	472	260	81	6	0	0	0	2	53	259	574
	HDD65	5256	1073	884	669	385	141	11	2	2	55	320	676	1038
	CDD50	3972	1	9	57	158	414	737	914	847	579	221	33	2
	CDD65	1435	0	0	1	12	85	298	450	384	182	23	0	0
	CDH74	17425	1	5	59	278	1094	3623	5590	4322	2085	352	16	0
	CDH80	8997	0	1	7	84	439	1877	3196	2297	980	114	2	0
Wind	WSAvg	11.0	9.9	11.0	12.1	13.1	11.7	11.8	10.6	9.8	10.9	11.2	10.6	9.9
Precipitation	PrecAvg	22.7	0.4	0.6	1.7	2.0	3.0	3.3	3.7	2.8	2.2	1.4	1.0	0.6
	PrecMax	38.0	1.7	2.7	7.9	5.3	9.0	7.7	11.6	7.1	8.5	3.8	3.7	2.1
	PrecMin	14.3	0.0	0.0	0.0	0.0	0.2	0.7	0.0	0.2	0.2	0.0	0.0	0.0
	PrecStd	5.1	0.4	0.6	1.7	1.1	1.7	2.0	2.3	1.8	1.7	1.1	1.0	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	72.6	80.6	88.2	96.8	104.4	106.1	102.5	99.6	90.2	76.0	64.2
		MCWB	46.9	51.3	57.5	60.5	66.4	70.4	71.3	71.0	69.3	65.2	55.6	48.9
	2%	DB	58.6	64.5	73.5	81.6	90.0	98.9	102.0	99.5	94.3	82.2	70.2	57.4
		MCWB	43.8	48.2	54.9	59.8	66.9	70.8	71.9	71.7	69.4	61.6	53.2	45.0
	5%	DB	53.3	57.4	69.0	75.5	84.2	94.5	99.2	96.7	90.3	77.3	64.0	53.6
		MCWB	41.4	44.9	52.9	58.1	65.5	70.8	72.1	71.8	68.8	60.0	50.4	43.0
	10%	DB	46.3	52.3	63.1	70.5	80.7	90.6	95.2	91.4	85.8	72.4	59.1	47.3
		MCWB	37.9	42.3	50.8	55.8	64.5	70.0	71.9	71.0	66.9	58.1	48.0	39.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	53.5	62.6	65.3	72.3	76.5	77.7	77.3	74.0	69.5	58.9	51.6
		MCDB	62.8	65.9	73.5	78.8	85.3	92.5	93.7	91.0	88.7	82.3	70.1	58.6
	2%	WB	44.9	49.8	58.8	62.4	70.1	74.4	75.6	75.3	72.0	65.5	55.0	46.3
		MCDB	57.1	62.8	67.0	75.5	83.4	90.5	91.5	90.0	87.9	76.5	65.3	56.1
	5%	WB	41.7	46.0	54.9	59.7	68.0	72.6	74.4	73.8	70.3	62.6	51.8	42.8
		MCDB	52.0	57.0	65.8	72.8	80.1	89.3	90.9	89.4	85.7	73.8	62.6	52.7
	10%	WB	38.6	42.4	50.9	57.0	65.9	71.1	73.2	72.4	68.5	59.4	48.7	39.3
		MCDB	46.4	52.0	63.2	69.5	77.4	87.4	90.1	87.9	82.0	70.1	59.1	46.8
Mean Daily Temperature Range	5% DB	MDBR	23.3	24.1	25.9	25.8	24.6	25.5	25.5	24.6	25.5	25.8	25.4	22.3
		MCDBR	34.0	34.8	35.7	34.9	31.7	30.8	31.2	30.6	30.1	33.4	33.9	31.5
		MCWBR	21.9	21.5	20.2	18.0	14.4	11.1	10.1	10.7	11.8	16.6	20.2	20.8
	5% WB	MCDBR	32.2	33.1	30.5	29.9	26.8	27.4	26.2	25.7	26.0	27.0	31.5	29.8
		MCWBR	21.3	21.1	18.5	16.9	14.3	11.9	10.2	10.7	11.6	15.2	19.3	20.3
Clear-Sky Solar Irradiance	taub	0.271	0.279	0.312	0.339	0.364	0.385	0.407	0.392	0.356	0.312	0.278	0.266	
	taud	2.532	2.514	2.439	2.392	2.404	2.377	2.351	2.386	2.422	2.532	2.550	2.538	
	Ebn at Noon	293	304	301	297	290	283	276	278	282	287	287	286	
	Edh at Noon	23	27	33	37	37	38	39	37	33	27	23	22	
All-Sky Solar Radiation	RadAvg	790	1046	1415	1750	2019	2256	2205	1945	1610	1174	874	692	
	RadStd	63	104	114	96	129	124	97	118	97	122	72	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.83	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page